



DEPARTMENT OF INFORMATION TECHNOLOGY
AND COMMUNICATION

DIPLOMA IN SOFTWARE DEVELOPMENT

PROJECT PROPOSAL

TITLE:

CASADIVE VILLA RESERVATION SYSTEM

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Table of Contents

1.0 INTRODUCTION.....	2
2.0 PROBLEM STATEMENT	3
3.0 OBJECTIVE	4
4.0 SCOPE	5
4.1 SYSTEM SCOPE.....	5
4.2 USER SCOPE.....	7
4.3 LOCATION SCOPE.....	8
5.0 PROJECT SIGNIFICANCE	9
6.0 LITERATURE REVIEW	10
6.1 INTRODUCTION.....	10
6.2 HOMESTAY PADI HIJAU	10
6.3 HOMESTAY CIKGU PAUH PUTRA.....	11
6.4 CASADIVE VILLA	12
6.6 DISCUSSION	13
6.7 CONCLUSION	13
7.0 METHODOLOGY	13
7.1 INITIAL REQUIREMENTS	14
7.2 SYSTEM DESIGN.....	15
7.3 PROTOTYPING.....	15
7.4 CUSTOMER EVALUATION	16
7.5 REVIEW AND UPDATION.....	16
7.6 DEVELOPMENT	17
7.7 TESTING.....	17
7.8 DEPLOYMENT	18
8.0 REFERENCES.....	18
10.0 COST PLANNING	19
11.0 CONCLUSION.....	20
12.0 APPENDIX A	21

1.0 INTRODUCTION

Casadive Villa was a homestay located in Kota Kuala Muda, Kedah, and it officially began its operations in April 2026. The villa was located beside the beach, making it an attractive destination for guests who wanted to enjoy a private and relaxing family vacation. Casadive Villa was selected as the case study for this Final Year Project (FYP) because the initial study showed that the company still managed its daily operations manually.

The study found that customer bookings, payment records, room maintenance, and customer information were managed through messaging applications and manual record keeping. This process caused slow operations and increased the risk of missing records, duplicate bookings, and data errors. In addition, customers had limited access to important information such as property details, house rules, nearby facilities, staff information, and the property's pet-friendly status. There was also no platform for customers to submit ratings or complaints, making it difficult for the management to collect customer feedback and improve service quality.

Furthermore, the management did not have an analytical system to monitor business performance. Information such as room for occupancy rates, monthly revenue, peak booking periods, and staff performance was not readily available, making business decision-making more difficult. Therefore, a web-based accommodation management system was proposed to digitalize and centralize the company's operations. The proposed system was expected to improve operational efficiency, enhance customer experience, and support better decision-making through real-time data and analytics.

2.0 PROBLEM STATEMENT

Casadive Villa manages its daily operations manually by using messaging applications and record files to handle customer bookings, payment records, room maintenance, and customer information. This method often resulted in duplicate bookings, missing records, and difficulties in monitoring room availability and status. As a result, the management process became slower, less organized, and more prone to human error.

In addition, Casadive Villa did not have a centralized system to support business management and customer services. Customers had limited access to important information such as house rules, nearby facilities, staff information, and the property's pet-friendly status. They were also unable to submit ratings or complaints, making it difficult for the management to collect feedback and improve service quality. Furthermore, the management could not efficiently manage staff information or monitor staff performance due to the absence of a proper staff management system.

Moreover, the management lacked an analytical system to monitor business performance. Important information such as room occupancy rates, monthly revenue, peak booking periods, and overall operational performance was not readily available. This made it difficult for the management to make accurate and timely business decisions. Therefore, a web-based accommodation management system was proposed to improve operational efficiency, enhance customer experience, and support better decision-making through real-time data and analytics.

3.0 OBJECTIVE

1. Develop a web platform to enable customers to view villa details, check availability, and make reservations seamlessly.
2. Integrate Power BI analytics to provide interactive dashboards and statistical insights on booking trends and customer data.
3. Implement automated notifications via email to send confirmations, and updates to enhance customer experience.

4.0 SCOPE

4.1 SYSTEM SCOPE

The proposed Villa Booking and Management System is a web-based system developed to improve and simplify the booking and management process at Casadive Villa, Kedah. The system provides customers with an easy, user-friendly, and elderly-friendly platform for booking accommodation while helping the management efficiently manage reservations, payments, and room information. The system consists of the following modules:

1. Room Details Management

- Add, update, and delete room information.
- Display room type, description, capacity, facilities, pricing, and images.
- Update room availability status.
- Manage seasonal room pricing.
- Allow customers to view complete room details before making a booking.

2. Booking Management

- Allow customers to book rooms online.
- Generate a unique booking number for each reservation.
- Check room availability based on selected check-in and check-out dates.
- Prevent duplicate bookings.
- Update booking status (Pending, Confirmed, Checked-in, Checked-out, Cancelled).
- Store customer booking history.

3. Customer Management

- Store customer information securely.
- Maintain customer booking records.
- Allow customers to view booking details.
- Manage customer profiles and contact information.

4. Report Management

- Generate booking reports.
- Generate payment reports.
- Generate room occupancy reports.
- Generate monthly revenue reports.
- Export reports in PDF or Excel format.

5. Business Analytics Dashboard (Power BI)

- Room Occupancy Rate.
- Monthly Revenue.
- Booking Trends.
- Peak Booking Seasons.
- Available vs Occupied Rooms.
- Popular Room Types.
- Booking Cancellation Analysis.
- Customer Statistics.

4.2 USER SCOPE

1. Administrator

- Manage room details (add, update, delete rooms).
- Manage room pricing and availability.
- Manage customer bookings.
- Monitor and verify online payment transactions.
- Generate booking and payment reports.
- Manage staff accounts.
- View business analytics dashboard.
- Manage customer records.

2. Staff

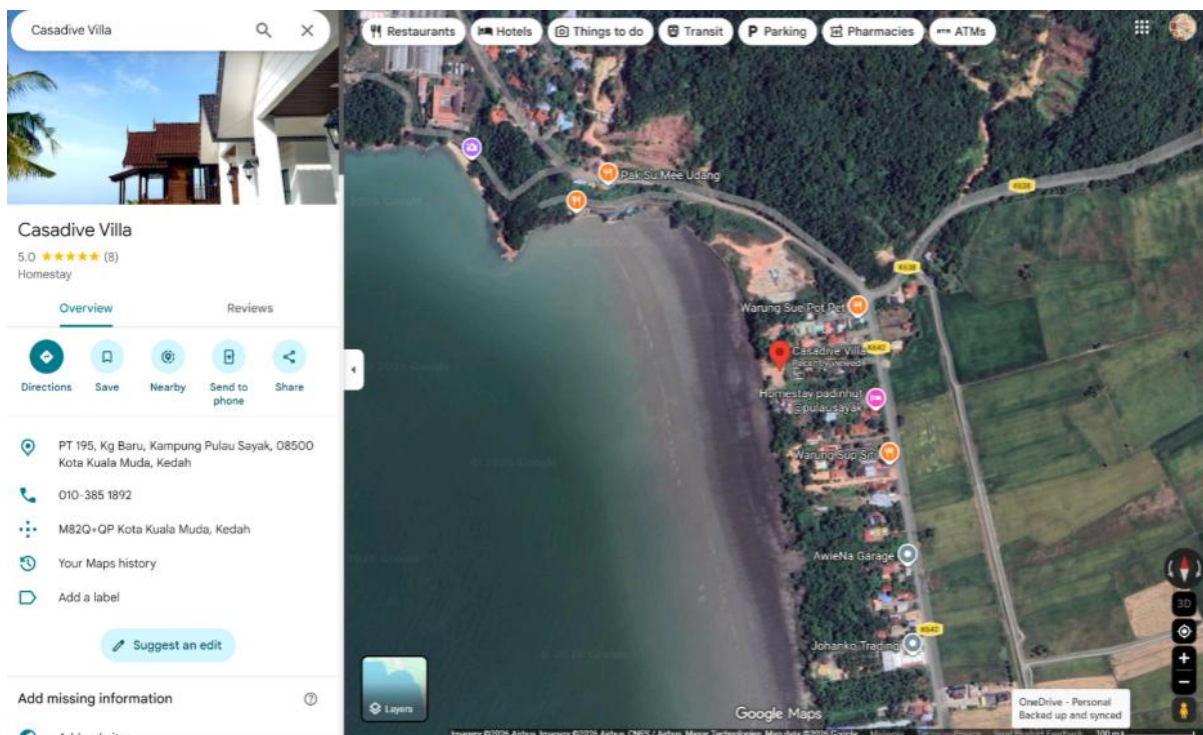
- View and manage customer bookings.
- Update booking status (Pending, Confirmed, Checked-in, Checked-out, Cancelled).
- Verify customer payment status.
- Assist customers with booking enquiries.
- Update room availability after check-in and check-out.
- Record room maintenance status.
- Generate booking confirmations and payment receipts.
- View customer booking information.

3. Customer

- Browse available rooms.
- View room details, facilities, pricing, and images.
- Check room availability.
- Make online bookings.
- Receive booking confirmation and digital payment receipt.
- View booking status using their booking reference.
- Contact Casadive Villa through the website for enquiries.

4.3 LOCATION SCOPE

The proposed system will be implemented at Casadive Villa, located in Kg Baru, Kampung Pulau Sayak, Kedah.



5.0 PROJECT SIGNIFICANCE

The development of the Casadive Villa Booking Management System provides several significant benefits that cannot be measured quantitatively but contribute greatly to the overall effectiveness of the system.

First, the system improves user convenience by allowing customers to access booking information and make reservations anytime and anywhere without needing to contact the villa management manually. This creates a more flexible and comfortable booking experience.

Besides that, the system helps improve management efficiency in handling room bookings, payment records, and room information. A centralized system allows villa management to organize data in a more systematic and structured way.

Moreover, the system reduces the possibility of human error. Manual booking processes may lead to issues such as double bookings, missing records, or inaccurate payment tracking. With an automated system, these risks can be minimized.

Lastly, this project supports digital transformation for Casadive Villa by shifting traditional manual operations into a more modern digital platform, improving the overall professionalism and service quality of the business.

6.0 LITERATURE REVIEW

6.1 INTRODUCTION

This section discusses the comparison between two existing homestay booking systems, both homestays currently use manual booking methods via WhatsApp or phone calls. This comparison is conducted to identify the limitations of existing systems and to justify the need for a more efficient, systematic, and web-based booking system.

6.2 HOMESTAY PADI HIJAU

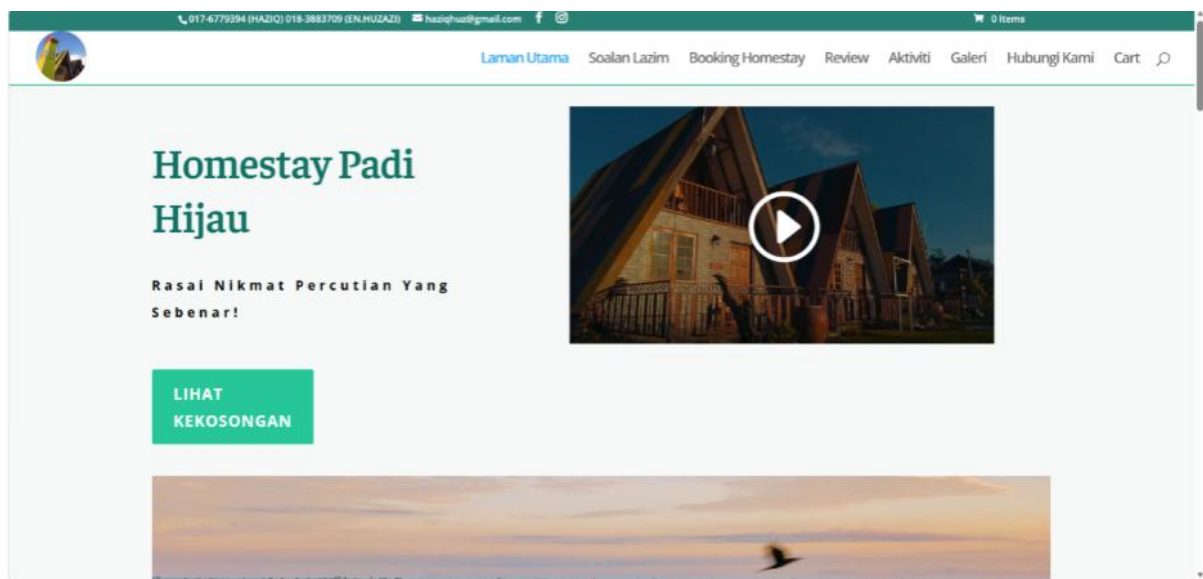


Figure 6.2 Homestay Padi Hijau

Homestay Padi Hijau operates using a manual booking approach where customers must contact the owner directly through WhatsApp or phone calls to make reservations. Room information is usually shared in the form of images or messages, without any centralized booking database.

Although this method is easy to manage and does not require system development costs, it has several limitations. The absence of an automated booking system increases the risk of double booking, and all reservation records are stored manually, making it difficult to manage and retrieve booking history efficiently.

6.3 HOMESTAY CIKGU PAUH PUTRA

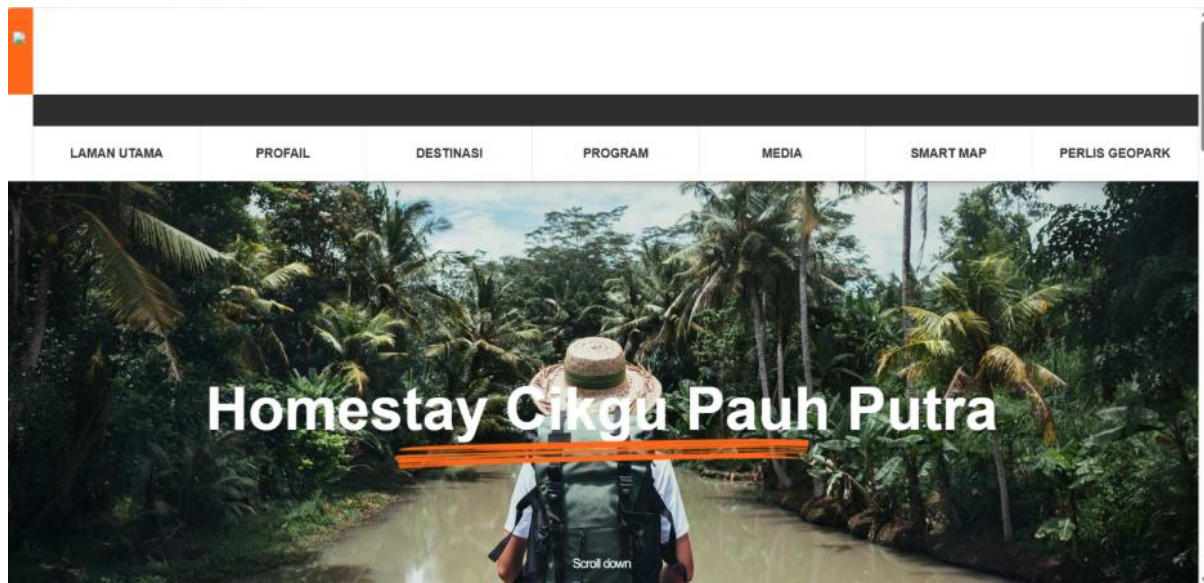


Figure 6.3 Homestay Cikgu Pauh Putra

Homestay Cikgu Pauh Putra also uses a similar manual booking system. Customers are required to contact the owner directly via WhatsApp or phone calls to confirm reservations, and all booking confirmations are handled manually.

While this approach is suitable for small-scale operations and allows direct communication between customers and the owner, it lacks proper data management. There is no centralized record system, no real-time availability checking, and it becomes inefficient when the number of bookings increases.

6.4 CASADIVE VILLA

The Casadive Villa Booking System is a web-based booking system designed to overcome the limitations of existing manual systems. It allows users to make online reservations, view room availability, and manage bookings through a centralized database.

This system automatically records all booking data, reducing the risk of double booking and human error. In addition, it provides real-time availability checking, making the booking process more efficient, systematic, and user-friendly for all users, including elderly users.

6.5 SYSTEM COMPARISON

Aspect	Homestay Padi Hijau	Homestay Cikgu Pauh Putra	Casadive Villa System
Booking method	WhatsApp	WhatsApp	Web-based online system
Database system	x	x	/
Booking records	x	x	/
Availability checking	x	x	/
Risk of double booking	high	high	/
Booking confirmation	manual	manual	automated
User convenience	/	/	/
Scalability	x	x	/

Table 6.5 Comparison Table

6.6 DISCUSSION

Based on the comparison above, both existing homestays rely heavily on manual booking systems. Although these methods are simple and cost-effective, they suffer from several weaknesses such as unorganized records, lack of centralized data storage, and high risk of booking conflicts.

In contrast, the proposed Casadive Villa Booking System provides a more modern and efficient solution. By implementing a web-based system with a centralized database, it improves data management, reduces human errors, and enhances the overall booking experience.

6.7 CONCLUSION

In conclusion, the existing manual booking systems used by both homestays are inefficient for long-term and scalable operations. Therefore, the development of the Casadive Villa Booking System is essential to provide a more systematic, reliable, and user-friendly booking platform.

7.0 METHODOLOGY

This project adopts the Prototyping Model as the software development methodology. The Prototyping Model allows the development team to create an initial prototype of the proposed system and obtain continuous feedback from the stakeholder before developing the final system. This methodology is suitable for the Casadive Villa Management System because it ensures that the proposed system meets the stakeholder's requirements while allowing improvements to be made throughout the development process.

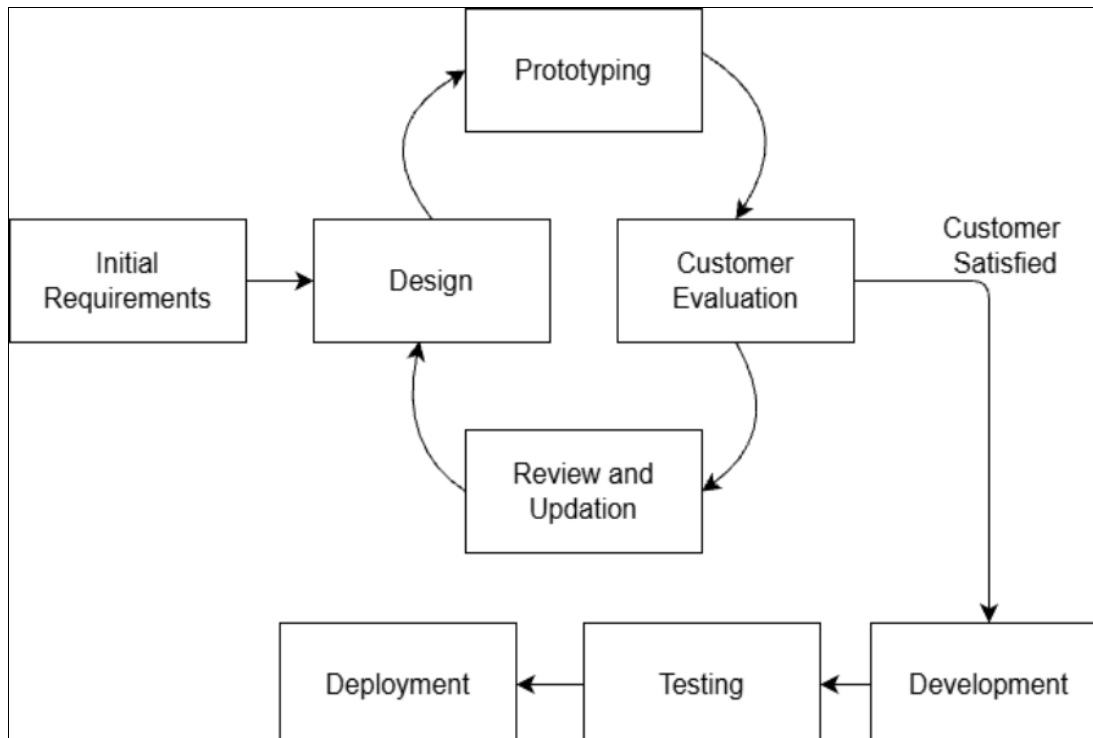


Figure 7.0 Prototyping Model

7.1 INITIAL REQUIREMENTS

This phase focuses on gathering and analysing the stakeholder's requirements. Information is collected through discussions and interviews with the owner of Casadive Villa to understand the current booking process, payment process, room management, and operational needs.

The activities carried out during this phase include:

- Conduct project planning.
- Gather information through interviews and discussions.
- Analyse the current booking process.
- Identify existing problems.
- Identify functional requirements.
- Identify non-functional requirements.
- Prepare the Software Requirement Specification (SRS).

7.2 SYSTEM DESIGN

This phase focuses on designing the overall structure of the proposed system based on the approved requirements. The system architecture, database, workflow, and user interface are designed before prototype development begins.

The activities carried out during this phase include:

- Design the Use Case Diagram.
- Design the Activity Diagram.
- Design the Entity Relationship Diagram (ERD).
- Design the Sequence Diagram.
- Design the Class Diagram.
- Design the database structure.
- Design the system navigation flow.
- Design the user interface (UI) using Figma.

7.3 PROTOTYPING

This phase focuses on developing an initial prototype of the proposed system to demonstrate its interface and main functionalities.

The activities carried out during this phase include:

- Develop the Login Page prototype.
- Develop the Customer Interface prototype.
- Develop the Staff Interface prototype.
- Develop the Administrator Interface prototype.
- Develop the Room Details prototype.
- Develop the Booking Module prototype.
- Develop the Payment Module prototype.

7.4 CUSTOMER EVALUATION

This phase focuses on evaluating the prototype with the stakeholder to ensure that it meets the operational requirements of Casadive Villa.

The activities carried out during this phase include:

- Demonstrate the prototype to the project supervisor.
- Demonstrate the prototype to the client.
- Collect user feedback.
- Evaluate the usability and functionality of the prototype.

7.5 REVIEW AND UPDATION

This phase focuses on reviewing the feedback received and improving the prototype before the final system development begins.

The activities carried out during this phase include:

- Analyse user feedback.
- Revise and improve the prototype.
- Update the system design where necessary.
- Obtain client approval before proceeding to the development phase.

7.6 DEVELOPMENT

This phase focuses on developing the complete web-based booking and management system based on the approved prototype.

The activities carried out during this phase include:

- Develop Elderly-Friendly User Interface
- Develop Login Module
- Develop the Room Details Module.
- Develop the Customer Module.
- Develop the Staff Module.
- Develop the Administrator Module.
- Develop the Booking Module.
- Develop Online Payment Module
- Integrate Payment Service
- Develop Reports Module
- Integrate the Power BI dashboard.
- Integrate all system modules.

7.7 TESTING

This phase focuses on verifying that the developed system functions correctly and satisfies the stakeholder's requirements.

The activities carried out during this phase include:

- Conduct Unit Testing.
- Conduct Integration Testing.
- Conduct System Testing.
- Conduct User Acceptance Testing (UAT).
- Fix identified errors and bugs.
- Validate the completed system.

7.8 DEPLOYMENT

This phase focuses on deploying the completed system for operational use by Casadive Villa. The system is configured and prepared to ensure that it can be used effectively by the stakeholder.

The activities carried out during this phase include:

- Deploy System to Server
- Configure the database.
- Configure Payment Service
- Conduct user training.
- Launch the system (System Go-Live)

8.0 REFERENCES

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10.0 COST PLANNING

Items		Cost	Quantity	Total Item Cost
Hardware				
1.	Laptop HP Pavilion 14	RM 3,199.90	1	RM 3, 199.90
2.	Laptop Asus Tuf Gaming A15	RM 3,999.00	1	RM 3,999.00
3.	Laptop Microsoft Surface	RM2,599.00	1	RM 2,599.00
4.	Mouse	RM30.00	3	RM 90.00
5	Pendrive	RM55.00	2	RM 110.00
Software				
1.	Microsoft Project Standard 2024 (Official Microsoft Store)	RM3,599.00	1	RM 3,599.00
2.	Visual Studio Code	RM 0.00	1	RM 0.00
3.	Xampp	RM 0.00	1	RM 0.00
4.	Domain name	RM 60.00	1	RM 60.00
5.	MySQL Server	RM 0.00	1	RM 0.00
6.	PHP	RM 0.00	1	RM 0.00
7.	Power BI Pro	RM 70.00	1	RM 350.00 / 5 unit
8.	Payment gateway (sandbox)	RM 0.00	1	RM 0.00
9.	Hosting	RM 180.00	1	RM 900.00 / 5 unit
10.	Github	RM 0.00	1	RM 0.00
11.	MySQL WorkBench	RM 0.00	1	RM 0.00
Other Requirements				
1.	Internet subscription	RM 120.00	3	RM 360.00
2.	Documentation printing	RM 20.00	20	RM 400.00
3.	Transportation (fuel & toll)	RM 100.00	5	RM 500.00
Total Overall Cost				RM 16, 166.90

Table 10.0 Cost Planning Table

11.0 CONCLUSION

In conclusion, the proposed web-based booking and management system for Casadive Villa is expected to provide a centralized platform to manage room reservations, payments, and room information more effectively. The proposed solution, which replaces existing manual processes with a web-based system, aims to improve record accuracy, reduce administrative workload, and provide customers with more convenient booking experience.

Furthermore, an elderly user interface will be a key component of the web-based system, incorporating larger font sizes, intuitive buttons, straightforward directions, and accessible formatting.

With Power BI, Casadive Villa can gain clear insights into booking trends, occupancy rates, payment status and revenue data through interactive dashboards and enabling management to make more informed business decisions.

The proposed system will be implemented through the Prototyping Model, which will allow stakeholders to give feedback along with the process. This approach ensures that the final system complies with Casadive Villa's operational needs and provides opportunities to improve interfaces and functionality before implementation.

Overall, the new system will improve operational efficiency and convenience for the customer through a more effective method of booking a room. It also provides a practical digital solution that supports the daily management activities of Casadive Villa.

12.0 APPENDIX A



POLITEKNIK TUANKU SYED SIRAJUDDIN
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KEPADA SESIAPA YANG BERKENAAN

Tuan / Puan,

KERJASAMA MEMBANTU PROJEK AKHIR PELAJAR BAGI KURSUS DFT50194 – INTEGRATED PROJECT

Dengan segala hormatnya perkara di atas adalah dirujuk.

2. Untuk makluman pihak tuan / puan, pelajar-pelajar seperti di **LAMPIRAN 1** adalah merupakan pelajar di Jabatan Teknologi Maklumat dan Komunikasi, Politeknik Tuanku Syed Sirajuddin, Perlis. Bagi memenuhi syarat penganugerahan Diploma Teknologi Maklumat (DIT), pelajar perlu membangunkan satu projek akhir melalui kursus DFT50194 – Integrated Project.

3. Sehubungan dengan itu, saya ingin memohon kerjasama dari pihak tuan / puan untuk memberi bantuan yang diperlukan oleh pelajar berkenaan dalam menyiapkan projek mereka bermula pada Jun 2026 hingga Oktober 2026. Sebarang masalah atau maklumat berhubung dengan perkara di atas, pihak tuan/puan boleh berhubung dengan pegawai berkenaan di talian 04-9886200 samb. 1318.

- i. Pn. Noor Halawati Binti Che Meh
- ii. En. Mohamad Izzat Bin Ismail

4. Segala bantuan dan kerjasama yang diberikan oleh pihak tuan / puan amatlah dihargai dan didahului dengan ucapan ribuan terima kasih.

Sekian.

"MALAYSIA MADANI"
"BERKHIDMAT UNTUK NEGARA"

Saya yang menjalankan amanah,

(HARYIAH BINTI RAHIM)
Ketua Jabatan Teknologi Maklumat dan Komunikasi
b.p. Pengarah
Politeknik Tuanku Syed Sirajuddin



